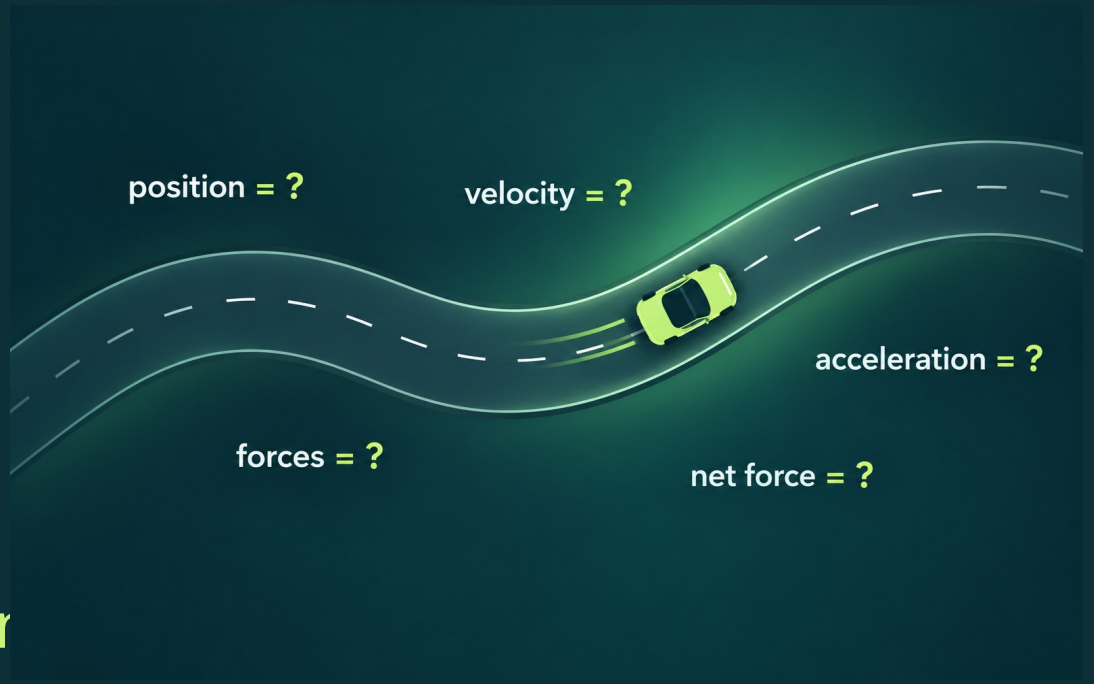
GRADE 10 SCIENCES / MYP 5

MOTION MAKES SENSE



Lesson 1 • Introducing Force and Motion

Observe · Describe · Explain · Predict · Control

ACTIVITY 1

Watch motion. Ask questions.

CAR-MOTION VIDEO MONTAGE

Acceleration • Cornering
Jumps • Braking • Crashes

WATCH VIDEO MONTAGE

Notice

What kinds of motion can you identify?

Wonder

What might influence or control that motion?

MYP CONCEPT: CHANGE

THINK • PAIR • SHARE

Describe the motion scientifically

Starting from rest

How does the motion begin?

Driving around a bend

What changes?

Jumping over a ramp

What happens to the path?

Coming to a stop

What might cause the change?

Use scientific language if you can. It is fine to be uncertain.

FROM OBSERVATION TO PHYSICS

Motion takes many forms

What did you see in the video?

CHANGING DIRECTION
Cornering

SLOWING DOWN
Braking

SPEEDING UP
Accelerating

FLYING THROUGH THE AIR
Jumping

TUMBLING
Rotating in a crash

Our starting point: simple one- and two-dimensional motion.

Full 3D motion and rotational motion come later, using the same foundations.

KEY POINT • Physics helps us describe, measure, predict and even control motion.

MYP CONCEPTUAL CONNECTION

Change, causes & consequences

CHANGE & CONSEQUENCES

Motion can look complicated in the real world. By building useful scientific models, we can describe changes in motion and explore what causes them.

We will connect these ideas to vehicles, everyday journeys and safety.

ORIENTATION IN SPACE AND TIME

Questions to keep asking

Where is it? How is it moving?
What changed? How do we know?
What happens next?

Our goal: observation → explanation, supported by measurements and evidence.

STATEMENT OF INQUIRY

Investigating the causality of change in motion allows us to understand our orientation in space and time, shaping our perception and interaction with the physical world.

THE PURPOSE OF PHYSICS

Complex motion → simple models

Physics helps us…

DESCRIBE
What is happening?

MEASURE
How much?

COMMUNICATE
How can we tell others?

PREDICT
What happens next?

EXPLAIN
Why does it happen?

CONTROL
How can we change it safely?

Models simplify reality. Their usefulness depends on the assumptions we make.

OUR FIRST LANGUAGE OF MOTION

Eight quantities we will learn to describe

01
Position

02
Distance travelled

03
Displacement

04
Speed

05
Velocity

06
Average speed

07
Average velocity

08
Acceleration

Today: clear meanings and examples. Later: mathematical definitions and calculations.

BUILDING BLOCKS

What is a physical quantity?

Quantity

Something measurable or describable quantitatively: e.g. position or speed.

Symbol

A short way of naming a quantity: e.g. x for position.

SI unit

An agreed measurement unit: e.g. metre (m).

20 m → a value + an SI unit

UNITS & SYMBOLS

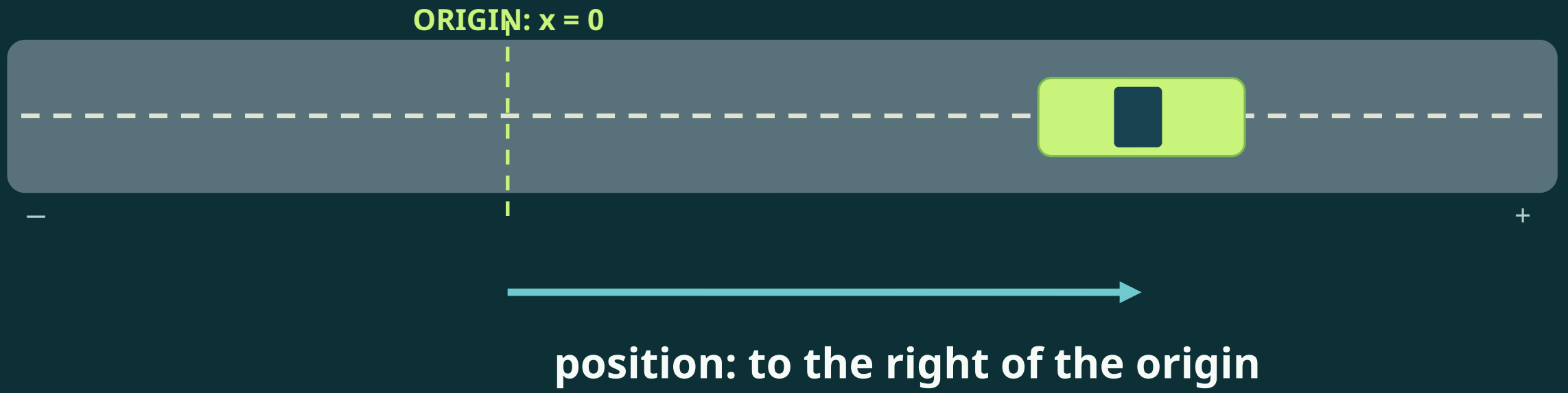
The quantities we will use

Quantity	Symbol (example)	SI unit
Position / distance / displacement	x / d / Δx	metre (m)
Speed / velocity	v	metre per second (m/s)
Average speed / average velocity	v_{avg} / $\bar{v}$	metre per second (m/s)
Acceleration	a	metre per second squared (m/s^2)

Δ means “change in”. Different books may use slightly different symbols: define yours clearly.

WHERE IS THE CAR?

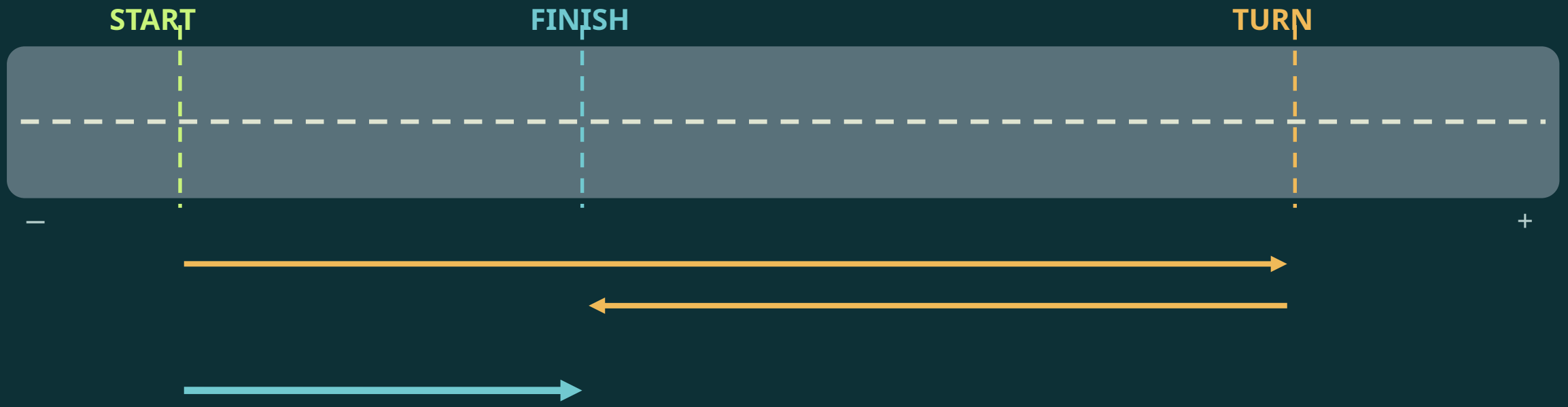
Position needs a reference point



Position is not meaningful until we choose a reference point.

A JOURNEY WITH A RETURN

Distance travelled ≠ displacement



DISTANCE: the entire path driven

DISPLACEMENT: start → finish

SPEED AND VELOCITY

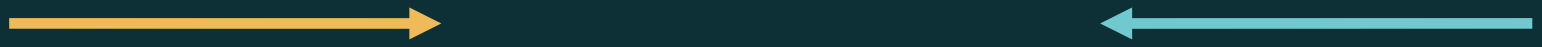
How fast? And in which direction?

SPEED

How fast a car is moving. Direction is not included.

VELOCITY

How fast the car is moving AND in which direction.



Same speed, different velocities

SNAPSHOT OR WHOLE JOURNEY?

Instantaneous vs average quantities

AT AN INSTANT

Position now
Speed now (speedometer)
Velocity now
Acceleration now

OVER A TIME INTERVAL

Distance travelled
Displacement
Average speed
Average velocity

The speedometer reading at one moment need not equal the average speed of the trip.

ACCELERATION IS NOT ALWAYS SPEEDING UP

Velocity changes in three ways

SPEED UP

Velocity magnitude increases.

SLOW DOWN

Velocity magnitude decreases.

TURN

Velocity direction changes.

Even at constant speed, a car turning a corner is accelerating.

MAGNITUDE AND DIRECTION

Scalars or vectors?

SCALAR

Size only
Distance · Speed · Average speed

VECTOR

Size and direction
Displacement · Velocity · Average velocity · Acceleration

In straight-line motion we choose right as + and left as −.

INTERACTIVE INVESTIGATION

Six car journeys — Observe first

Only analyse the time while the screen is GREEN.

1

Select journey A–F

2

Press GO

3

Replay and discuss

4

Reveal the interpretation

Ask: what happens at an instant, and what happens over the entire interval?

DISCUSSION ANSWERS

What do journeys A–D show?

A · Constant velocity

Same speed and same direction. Acceleration is zero.

B · Circular journey

Speed constant; velocity changes direction. After one lap, displacement = 0.
(Acceleration always points towards the centre of the circle.)

C · Accelerating

Starts at rest and speeds up uniformly.
Acceleration is in the same direction as its velocity.

D · Braking

Starts fast and slows uniformly to rest.
Acceleration is in the opposite direction to its velocity.

C and D cover the same distance in the same time but have different speed patterns.

DISCUSSION ANSWERS

What do journeys E–F show?

E · Moving left

Speed is positive, but velocity is negative when right is chosen as positive.

F · Return journey

The car reverses and returns. Distance is non-zero; displacement and average velocity are zero. Acceleration is always left (negative direction).

One big idea: a car can move throughout an interval yet have zero average velocity.

HOMEWORK • EXPLORE

Running Car: predict, then press GO

STARTING POSITION

Where does the car begin relative to the origin?

INITIAL VELOCITY

Which direction, and how fast at the start?

ACCELERATION

Which way is velocity changing?

Make a prediction → GO → Observe → RESET → Try again

Make notes on anything you have learned or discovered.

EXIT TICKET

Four questions to take away

01 What does velocity tell us that speed does not?

02 Can a car accelerate without speeding up?

03 Can a car travel a distance but have zero displacement?

04 What do positive and negative values of position, velocity and acceleration mean?

MYP: ORIENTATION IN SPACE & TIME

Next: move from everyday descriptions towards precise measurements.

WHAT WE HAVE LEARNED

Eight quantities for describing motion

POSITION
Where the car is relative to a chosen origin.

DISTANCE TRAVELLED
How much ground the car covers along its path.

DISPLACEMENT
Change in position, including direction.

SPEED
How fast the car is moving.

VELOCITY
How fast it moves and in which direction.

AVERAGE SPEED
How fast the car travels overall during a journey.

AVERAGE VELOCITY
How quickly position changes overall, including direction.

ACCELERATION
How quickly velocity changes in size or direction.

These are the words we will turn into measurements and equations in Lesson 2.